SMBC Data & AI Workshop · Cheat sheet

Lakeflow, AI/BI, Genie Agents and Genie Code

All data is synthetic

1 Sign in

QR code for the workspace

<workspace URL>

Microsoft Edge or Google Chrome; sign in with your SMBC account.

2 Check in

SQL Editor → the workshop's warehouse → paste, **Run** (`'DE'` for engineers). It shows your Lab 1 schema.

```
BEGIN
  DECLARE my_schema STRING DEFAULT
      regexp_replace(lower(split_part(current_user(),
      '@', 1)), '[^a-z0-9_]', '_');
  INSERT INTO smbc_workshop.ops.checkin
  SELECT current_user(), 'DA', current_timestamp();
  CREATE SCHEMA IF NOT EXISTS
      IDENTIFIER('smbc_workshop.' || my_schema);
  SELECT 'smbc_workshop.' || my_schema AS your_schema;
END
```

Lab 1 · Build the trusted exposure table

1. **+ New** → **Visual data prep**. Name it *Monday exposure report*; compute: **Serverless**.
2. **Sources** from `smbc_workshop.raw.landing`: `counterparties`, `facilities`, `monthly_balances`, `fx_rates`, and the limits (`group_limits.xlsx` or its folder).
3. **Unique** on counterparties, all columns: 2,370 → 2,325 rows.
4. **Genie Code** on facilities: clean `currency`, parse both date formats (prompt in A.2).
5. **Guardrails**: `committed_amount` > 0, `maturity_date >= start_date`, dates not null. **Failed** → Output `quarantine_facilities`: 21 failed rows.
6. **Balances**: parse `as_of_date`, add `month`; **Unique** on `facility_id`, `as_of_date`, latest `extract_ts`: 74,799 → 74,757. **FX**: add `rate_month`.
7. **Join** with Genie Code (A.2): balances → facilities → counterparties → FX on currency and month; add `exposure_usd`, `exposure_jpy`.
8. **Aggregate** by `as_of_date`, `parent_group_id`; **Join** the limits; `utilisation_pct` = 100 × exposure ÷ limit.
9. **Output** `gold_group_exposure` (Overwrite) → **Run** → **Schedule** daily, 07:00 Singapore time.

✅ **Checkpoint:** the two totals should be identical.

```
SELECT round(sum(exposure_usd) / 1e6, 1) AS mine_usd_m
FROM smbc_workshop.<your_schema>.gold_group_exposure
WHERE as_of_date = '2026-09-30';
SELECT round(sum(exposure_usd) / 1e6, 1)
    AS reference_usd_m
FROM smbc_workshop.gold.group_exposure
WHERE as_of_date = '2026-09-30';
```

Lab 2 · Credit Committee in 30 seconds

1. **Dashboard**: + New → Dashboard on both `gold` tables. Ask Genie Code for the top-10 bar chart; add counters, a monthly trend and exposure by country; filters; **Publish**. Then open `<workspace URL>/one`.
2. **Genie Agent**: Genie Agents → New on the same two tables. Title *Credit Risk – <your name>*, the warehouse you checked in with; add the instructions (A.3), example SQL (A.5) and questions (A.6).
3. **Ask, then open the SQL.** Trust, but verify.
4. **Fiscal-year test**: ask about Q1 FY2026 and FY2025 and check the dates in the SQL. Add A.4, ask again.

✅ **Checkpoint:** ask "*Which groups were above 90% of their limit at the end of FY2026 H1?*", then submit the group's name:

```
INSERT INTO smbc_workshop.ops.lab_answers
SELECT current_user(), 'lab2', '<group name>',
    current_timestamp();
```

Genie Code · Ask why, safely

> **Explore:** "Profile `smbc_workshop.gold.group_exposure` and tell me what is unusual about the latest month compared with April 2026."

> **Generate:** "For the group with the highest utilisation at 30 Sep 2026, split its change in exposure since April into an FX effect and a new-drawdowns effect, by subsidiary. Chart it."

> **Explain and convert:** "Explain this procedure and point out any risks, then rewrite it as Databricks SQL. Just explain first, don't run any code." Paste the legacy procedure from A.9.

Full prompts: workbook A.8, or notebook 03 in the Labs folder.

Getting help

Red: I need help | **Green: all good**

Stuck or short on time? Carry on from the reference tables in `smbc_workshop.gold`: nothing later depends on finishing Lab 1. Common problems and fixes: workbook Appendix B.

Keep learning

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